

Citation-Network Cohesion and Reciprocal Citation Cliques Across Journal-Impact Groups: A Matched Author Analysis

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Abstract

We ask whether authors in lower-impact journal groups have different citation patterns from matched authors in higher-impact groups. We call the lower-impact observations Cases and the higher-impact observations Controls. We compare 102,626 author–subject pairs within subject and at similar recent citation productivity. All four primary mean differences favored Cases; all four comparisons were significant after adjustment, and 3 median differences were zero. Primary-rule reciprocal-clique membership was 10.96% for Cases and 3.82% for Controls (paired difference 7.14 percentage points; exact $p < 0.001$). Lower-impact-group authors were more likely than matched higher-impact-group authors to belong to formally defined reciprocal citation cliques.

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Keywords: scientometrics; citation networks; matched design; network cohesion; journal impact; anomaly screening

1 Introduction

15 Citation indicators are widely used to summarize scholarly influence, allocate resources, and support evaluation. Their convenience also creates incentives to optimize what is measured, which can weaken the connection between an indicator and the quality it is intended to represent (Biagioli et al., 2019; Goodhart, 1984; Hicks et al., 2015). Excessive self-citation, coercive citation, and concentrated citation exchange are therefore important objects of
20 measurement in their own right (Biagioli & Lippman, 2020; Van Noorden & Chawla, 2019; Wilhite & Fong, 2012).

Most evidence on unusual citation structure is either journal-level or begins with groups already selected for unusual behavior. Journal-network studies can identify exceptional exchange patterns (Kojaku et al., 2021), and analyses of citation stacking show how journal-
25 level citation flows can be distorted (Heneberg, 2016). Author-level concentration indicators can show when a small set of collaborators accounts for much of an author’s citation impact (Evdaimon, Ioannidis, et al., 2024). Less is known about whether simple author-network measures differ across journal-impact groups after authors are compared within subject and at similar levels of recent citation productivity.

30 This study addresses that gap with a matched author–subject design. Low-impact and high-impact refer only to groups defined by subject-specific, publication-weighted 25th and 75th percentile cutoffs of journal Eigenfactor scores. We use *Case* and *Control* as compact labels for those operational groups; neither label evaluates the scientific merit of an author or work.

35 The study asks three questions:

1. Do matched Case and Control authors differ in citation-network cohesion, outgoing citation concentration, and reciprocal-clique membership?
2. Are author–subject profiles that are unusual relative to Controls enriched among Cases?
3. How do fractional citation weights mix across the two groups?

We expect Case author–subjects to have higher values than Controls on four primary 40
measures: citations to previous collaborators, reciprocal citation, local neighbor connections,
and concentration of outgoing citations. This is an associational hypothesis, not a causal
or integrity claim. Because many values are zero, a zero median shift does not rule out a
directional difference. Network closure can reflect legitimate specialization and collaboration,
so the anomaly procedure is a screening device. 45

2 Related Work

Research evaluation based on citation counts, the h -index, and journal indicators has long
been accompanied by cautions about field differences and strategic responses (Garfield, 2006;
Hicks et al., 2015; Hirsch, 2005). Self-citation is common and can be substantively appropriate,
but extreme rates complicate comparisons (Aksnes, 2003; Van Noorden & Chawla, 2019). 50
Coercive citation and metric gaming demonstrate that observed references can also respond
to editorial and institutional incentives (Baccini et al., 2019; Wilhite & Fong, 2012).

Several strands of work motivate a network perspective. Kojaku et al. identify anomalous
groups in journal citation networks by comparing observed patterns with network expectations
(Kojaku et al., 2021). Heneberg examines the transition from journal self-citation to citation 55
stacking through journal-level citation flows (Heneberg, 2016). These approaches establish
the value of relational structure but operate at the venue level. At the author level, Evdaimon
et al. propose indicators based on the concentration of citing authors and hyper-collaboration
(Evdaimon, Ioannidis, et al., 2024). Their concentration measures identify a narrow citing
base; local closure, reciprocity, and concentration of an author’s own references provide 60
complementary information about how that base is connected.

The present analysis combines this network perspective with within-subject author match-
ing. It does not attempt to classify behavior as legitimate or illegitimate. Instead, it
estimates matched differences in a small, declared set of network measures and applies a

65 Control-referenced screen whose output remains explicitly descriptive.

3 Data and Cohort

3.1 Source data and study window

The source is a fixed Crossref snapshot processed with *alexandria3k* (Spinellis, 2023). The study window covers citing and cited works dated 2020–2024. Available fields include
70 Digital Object Identifiers (DOIs), publication year, normalized journal International Standard Serial Numbers (ISSNs), work references, and author rows with optional ORCID researcher identifiers. ORCID is a persistent identifier for a researcher. The estimates apply to this fixed snapshot and to the author–subject observations that meet the stated inclusion criteria.

Journals are assigned to Health Sciences, Life Sciences, Physical Sciences, Social Sciences
75 and Humanities, or Multidisciplinary. The existing classification uses a deterministic, cached large-language-model annotation of representative journal content.

The retained venue classification uses fixed subject-specific journal scores from a separate directed journal-citation calculation (Bergstrom et al., 2008). An Eigenfactor score is a journal influence score based on citation flows. In plain terms, the calculation passes influence
80 from citing journals to cited journals, excludes journal self-citations, and uses article counts to redistribute influence from journals without outgoing citations and to restart the calculation. These journal-level scores define the two venue groups; they are an input to the comparison, not an author-network outcome. The exact matrix specification is given in Appendix A.

The current workflow consumes the fixed database scores as an upstream exposure rather
85 than re-estimating them. After scores are attached to eligible 2020–2024 work rows, the 25th and 75th percentile cutoffs are calculated across positive-score work rows within each subject. A journal therefore contributes once per eligible work: the cutoffs are publication weighted, not percentiles over distinct journals. Journals at or below the lower cutoff form the low-impact group and those at or above the upper cutoff form the high-impact group.

3.2 Author–subject cohort

The unit used throughout is an *author–subject*, meaning one researcher considered within one subject area, identified by (ORCID, s). An author is a Case in subject s when at least 70% of eligible works in that subject appear in low-impact journals, and a Control when at least 70% appear in high-impact journals. Authors not meeting either rule are excluded from matching. 95

A uniqueness check prevents more than one group assignment for the same (ORCID, s) key.

The primary cohort contains 102,626 matched Case–Control pairs across the five subjects. It is built from every non-empty ORCID in the fixed snapshot that meets the portfolio, work-count, and positive- h_5 eligibility rules; no final-character sampling rule is applied.

4 Matching

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Recent subject-specific citation productivity is summarized by h_5 , the Hirsch index calculated from an author’s works in subject s published during 2020–2024 (Hirsch, 2005). In ordinary language, h_5 is the largest number of works that each have at least that many citations in the fixed snapshot. The matching pool requires $h_5 \geq 1$ and excludes otherwise group-eligible zero- h_5 profiles before candidate generation. 105

We pair lower-impact and higher-impact author–subjects within the same subject when their h_5 values differ by at most three points. Candidates are considered in the fixed order subject, smallest absolute h_5 difference, Case ORCID, and Control ORCID; each author–subject can be used only once.

An independent audit repeats the deterministic match from the generated full-cohort pro-110 files and compares it with the materialized pairs. It requires identical order and membership, no pair beyond the three-point limit, and no reused author–subject key. The complete audit accompanies the replication materials.

Table 1: Matched authors by subject.

Subject	Pairs	h_5 observed	Exact h_5	Median Δh_5	Median $ \Delta h_5 $	Maximum $ \Delta h_5 $
Overall	102,626	102,626	64,176	0.00	0.00	2
1	40,439	40,439	17,869	-1.00	1.00	2
2	28,045	28,045	15,205	0.00	0.00	2
3	3,383	3,383	3,345	0.00	0.00	1
4	24,879	24,879	21,877	0.00	0.00	1
5	5,880	5,880	5,880	0.00	0.00	0

Δh_5 is Case minus Control. Every pair must satisfy $|\Delta h_5| \leq 3$. Exact- h_5 pairs form a sensitivity cohort; they do not replace the primary matched cohort.

The matching table supplies the Case ORCID, Control ORCID, subject, and both h_5 values.

115 Feature rows are linked back to this table twice, on $(\text{case_orcid}, s)$ and $(\text{control_orcid}, s)$; ORCID-only joins are not used.

We report the distribution of h_5 and the within-pair difference before outcome testing. Because residual differences can remain under caliper matching, the four primary comparisons are repeated on the subset with exact h_5 equality.

120 The generated cohort audit found a maximum observed absolute h_5 difference of 2 and 0 pairs beyond the three-point caliper.

5 Network Construction and Definitions

We represent authors as nodes and citations as links between nodes. A link has a direction, from the citing author to the cited author, and a fractional weight that accounts for the
125 number of authors on both works. We use the word “network” for this set of nodes and weighted links.

5.1 Reference resolution and fractional weights

A citing-work/cited-work DOI pair is deduplicated before either work is expanded to author rows. Let n_c and n_a be the complete counts of author rows on the citing and cited work.

These denominators include authors without ORCIDs. A resolved work citation contributes 130

$$\frac{1}{n_c n_a} \tag{1}$$

to each identified citing-author/identified cited-author combination. Rows with an unidentified endpoint do not produce analytic author edges; in particular, missing identifiers cannot become a shared pseudo-author. Multiple matched authors on one citing work do not multiply the underlying resolved work reference.

Every analytic edge is keyed by subject, citing ORCID, cited ORCID, and citation year. 135 Only citing works assigned to the matched author’s subject contribute to that author–subject’s measures. Duplicate rows are aggregated so that this key is unique.

For coauthor classification, let c_{ij} be the first year in which authors i and j published together. A citation from i to j in year t is a prior-coauthor citation only if $c_{ij} < t$. A sensitivity analysis uses $c_{ij} \leq t$ to show the effect of admitting same-year collaborations. 140

5.2 Graph and reference populations

The analysis uses three views of the citation links so that each measure has a clear population.

Each author’s outgoing links. For each matched author–subject i , this view contains citations from i to any identified author reached from an eligible citing work. It defines self-citation, citations to previous collaborators, outgoing concentration, and the largest 145 year-to-year increase to one recipient.

Links among matched authors. For each subject s , this view retains only matched authors and the links between them. It defines reciprocal citation, local neighbor connections, the balance of incoming and outgoing weight, and citation mixing between the two groups. 150

Deduplicated work references. This view is built before author expansion and defines

same-journal referencing even when the cited author is not identified.

Keeping these views separate prevents a measure based on all identified recipients or work references from being mistaken for a measure based only on matched authors.

155 **5.3 Notation, metrics, and missing values**

Some measures follow an author’s outgoing citations to any identified recipient; others use only links between matched authors. Direction is ignored only for the simple neighbor graph used for local clustering.

Let x_{ijt} denote the fractional citation weight from author i to author j in year t , after the
 160 construction above. Missing dyad-years are zero, and $w_{ij} = \sum_{t=2020}^{2024} x_{ijt}$ is cumulative weight. If V_s is the set of matched authors in subject s , the matched-author graph is the directed, weighted, loopless graph

$$G_s = (V_s, E_s, w), \quad E_s = \{(i, j) : i, j \in V_s, i \neq j, w_{ij} > 0\}. \quad (2)$$

Its simple binary undirected projection is

$$U_s = (V_s, \tilde{E}_s), \quad \{i, j\} \in \tilde{E}_s \iff i \neq j \text{ and } w_{ij} + w_{ji} > 0. \quad (3)$$

Direction and fractional weight are discarded only in U_s . Self-edges remain available in the
 165 outgoing ego layer for self-citation and surge share but do not enter G_s or its peer-network measures.

5.4 Formal reciprocal-clique analysis

We test citation-clique structure on all matched authors, not only on authors flagged by the screen. A clique C is a group of at least four matched authors in which every pair is
 170 connected by a citation in at least one direction. We keep only maximal groups, meaning that

no additional matched author can be added without breaking that rule. The direction and weight of the links are then evaluated separately. For a clique of size k , directed density is

$$d_C = \frac{m_C}{k(k-1)}, \quad (4)$$

where m_C is the number of positive directed dyads among the k members. Weighted reciprocity is

$$r_C = \frac{\sum_{\{i,j\} \subset C} \min(w_{ij}, w_{ji})}{\sum_{\{i,j\} \subset C} \max(w_{ij}, w_{ji})}. \quad (5)$$

The primary rule requires at least 75% of all possible directed links and a weighted reciprocity of at least 0.50. We repeat it for minimum clique sizes $k = 3, 4, 5$ and reciprocity thresholds 0.25, 0.50, 0.75; the density threshold remains fixed at 0.75.

For inference, overlapping qualifying cliques are converted to one binary membership value per author–subject. Case and Control membership rates are compared within the original matched pairs. The exact two-sided null distribution conditions on the number of discordant pairs and assigns either member of each discordant pair equal probability of carrying the clique-membership label. We report both discordant-pair counts, the paired rate difference, the exact p -value, and the direction of the rate difference within each subject.

The study-specific temporal measure is

$$B_i = \frac{\max_{j,t=2021,\dots,2024} [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}. \quad (6)$$

This study-specific temporal increment is not a state-transition burst-detection model. 185

Unavailable values remain missing through feature assembly. A zero is used only when the denominator and graph population are observed but the relevant event is absent. Thus a defined rate with no qualifying citations is zero; reciprocity is zero when internal outgoing weight exists but none is returned; clustering is zero for an observed induced-graph node with fewer than two neighbors; and journal endogamy is zero when resolved references exist 190

Table 2: Declared author–subject measures.

Measure	Definition	Population and interpretation
Prior-coauthor citation rate	$\frac{\sum_{t,j \neq i} x_{ijt} \mathbf{1}(c_{ij} < t)}{\sum_{t,j \neq i} x_{ijt}}$	Outgoing ego layer. Fraction of identified non-self outgoing weight directed to an author whose first collaboration preceded the citation year.
Self-citation rate	$\frac{\sum_t x_{iit}}{\sum_{t,j} x_{ijt}}$	Outgoing ego layer. Fraction of identified outgoing weight assigned to the same ORCID.
Reciprocity	$\frac{\sum_{j \neq i} \min(w_{ij}, w_{ji})}{\sum_{j \neq i} w_{ij}}$	Directed, weighted G_s . Among matched peers, the share of outgoing weight that is also returned to the author.
Local clustering	$\frac{2T_i}{k_i(k_i - 1)}$	Binary projection U_s . Among an author’s matched neighbors, the share of neighbor pairs that are connected to each other. Here k_i is the number of matched neighbors and T_i is the number of connected pairs among those neighbors.
Outgoing Herfindahl–Hirschman index (HHI)	$\sum_{j \neq i} p_{ij}^2, p_{ij} = w_{ij} / \sum_{k \neq i} w_{ik}$	Outgoing ego layer. Concentration of identified non-self outgoing weight across recipients; values near 1 mean that one recipient receives nearly all of the weight (Hirschman, 1964).
Journal endogamy	Resolved outgoing work references whose citing and cited journals have the same nonblank normalized ISSN, divided by all resolved outgoing work references	Deduplicated work-reference layer. Share of resolved references whose citing and cited journals have the same normalized ISSN. This descriptive term does not imply impropriety.
Within-sample citation balance	$\frac{\sum_{j \neq i} w_{ij} - \sum_{j \neq i} w_{ji}}{\sum_{j \neq i} w_{ij} + \sum_{j \neq i} w_{ji}}$	Directed, weighted G_s . Positive values mean more outgoing than incoming weight among matched peers; negative values mean more incoming weight.
Maximum annual dyadic surge share	$\frac{\max_{t=2021, \dots, 2024} \sum_j [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}$	Outgoing ego layer. Largest year-to-year increase to one recipient, divided by all identified outgoing weight.

Notes. x_{ijt} is fractional citation weight, w_{ij} is cumulative weight, c_{ij} is the first joint-publication year, and $\mathbf{1}(\cdot)$ is an indicator. G_s is the directed matched-author graph, U_s is its binary neighbor graph, and $[z]_+ = \max(z, 0)$. Measures are calculated separately for each author–subject; the zero and missing-value rules follow the table.

but none remains within the citing journal. Outgoing HHI is missing when there is no identified non-self outgoing weight, balance is missing when total internal incident weight is zero, journal endogamy is missing when no resolved outgoing work reference exists, and surge share is missing when total identified outgoing weight is zero. These rules imply $0 \leq B_i \leq 1$. Statistical comparisons use complete-pair deletion separately for each metric rather than a 195 common imputed matrix.

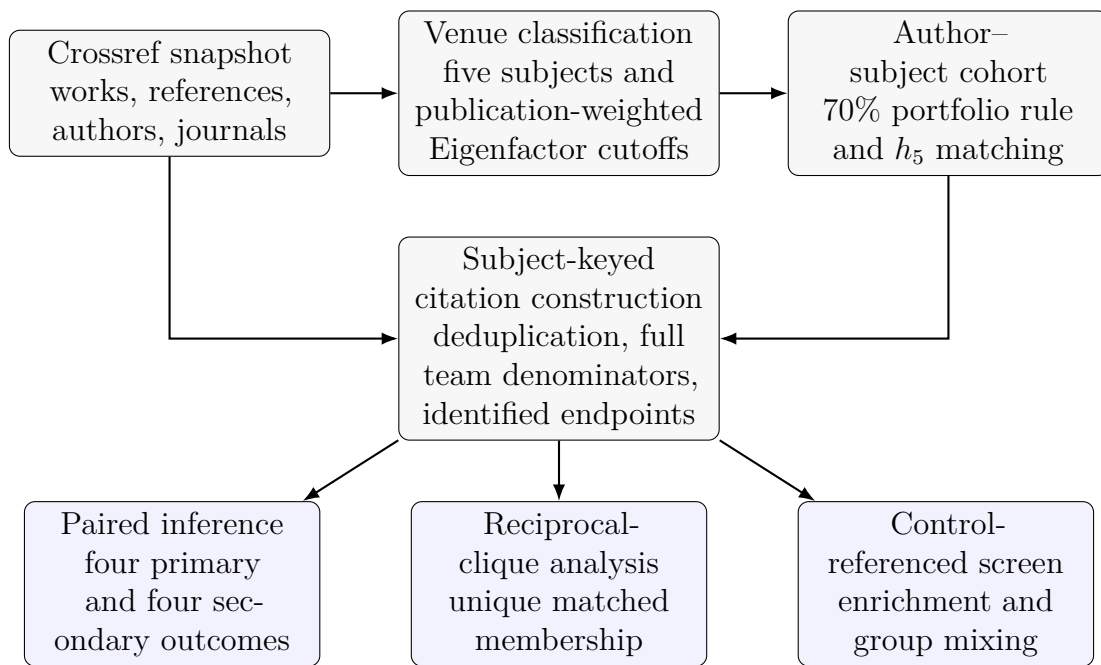


Figure 1: Reader-facing study design. Venue classification and author matching define the cohort; subject-keyed citation construction then supports the paired, clique, and screening analyses.

6 Paired Inference

The four primary outcomes are citations to previous collaborators, reciprocal citation, connections among an author’s citation neighbors, and outgoing HHI. HHI is a concentration score from 0 to 1; it is higher when an author sends most outgoing citation weight to a small number of recipients. The four secondary outcomes are self-citation, same-journal referencing, the balance of incoming and outgoing links, and the largest year-to-year increase to one recipient. The former normalised burst-intensity feature is not an additional outcome; it was replaced by the surge share in Equation 6. The revised analysis therefore reports eight distinct outcomes in two declared families. 205

For each matched pair, we subtract the Control value from the Case value. A positive difference therefore means that the Case value is larger. Only pairs with both values available enter a comparison. Because the measures contain many ties and zeros, we use the two-sided paired Wilcoxon signed-rank test, which does not assume a normal distribution (Wilcoxon, 1945). We report the number of pairs, the median difference, and a rank effect size: 210

$$r_{\text{rb}} = \frac{W^+ - W^-}{W^+ + W^-}, \quad (7)$$

where W^+ and W^- add the ranks of positive and negative differences after zero differences are omitted. The effect ranges from -1 to 1 ; a positive value means that Case ranks tend to be higher. We obtain 95% intervals from 2,000 bootstrap samples of matched pairs (Efron & Tibshirani, 1993). We also report mean differences and their bootstrap intervals because they give an intuitive measure of the size of the shift. Benjamini–Hochberg correction adjusts the 215 four p -values in each outcome family for the number of tests (Benjamini & Hochberg, 1995).

As a randomization check, we randomly reverse the sign of each pair difference in 10,000 runs and compare the resulting mean differences with the observed one. This keeps the size of every pair difference while removing its direction. We also repeat the primary comparisons for exact- h_5 matches and for a definition that counts collaborations formed in the citation 220

year.

7 Anomaly Screen

The anomaly analysis asks a different question from the matched comparison: are unusual profiles more common among Cases than among Controls? It is a screening rule, not a
225 prediction model and not evidence of intent.

An author–subject enters the screen only when it has at least one identified non-self outgoing citation. The screen uses five measures of an author’s citation links: self-reference, citations to previous collaborators, mutual exchange, connections among neighbors, and concentration among recipients. These inputs capture author-edge cohesion or concentration.
230 Same-journal referencing is excluded because it is defined on work references, citation balance because it requires internal incident weight, and year-to-year increase because it is temporal; all three are reported separately. All five screen inputs must be observed, and an unavailable value is not changed to zero.

Within each subject, every screen input is centered at the Control median and divided
235 by the Control interquartile range (IQR). A nonfinite or zero Control IQR is replaced by 1 only to avoid division by zero. Controls therefore define the preprocessing, detector fit, and thresholds; Cases do not affect those reference quantities.

No additional feature weights are applied, and the empirical percentile threshold makes no Gaussian assumption.

240 An Isolation Forest is a tree-based method that gives a higher score to a profile that is easy to separate from typical Controls (F. T. Liu et al., 2008). A row passes this part of the screen when its score is above the 99th percentile of Controls in the same subject. A row also passes the second part when at least two of four cohesion measures exceed their Control percentiles. The final flag requires both parts. Because the four cohesion measures also enter
245 the first part, the two parts are related and are not independent evidence.

Sensitivity analyses vary the Control percentile, random seed, and detector feature set. We report overlap between the detector and cohesion restriction, their rank association, and leave-one-feature-out stability. These quantities describe an operational screen; they are not measures of classification accuracy because group is not a ground-truth anomaly label.

Finally, we summarize where citation weight goes. The 2×2 table has the citing group in 250 the rows and the cited group in the columns. Weighted assortativity compares the observed share of same-group citations with the share expected from the row and column totals (Newman, 2002):

$$r = \frac{\text{tr}(e) - a^\top b}{1 - a^\top b}. \quad (8)$$

Here e is the observed 2×2 table of citation-weight shares, a contains its row totals, b contains its column totals, and $\text{tr}(e)$ is the observed same-group share. Positive r indicates 255 more same-group citation weight than expected from those totals. The reported label-swap p -value compares the observed same-group share with 10,000 runs in which the two labels are swapped within each matched pair. The swaps keep the citation links and matching structure unchanged while removing the original group labels.

8 Results

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All numerical values in this section come from the same analysis and are shown in the tables and figures.

8.1 Matched differences in network measures

The primary and secondary matched comparisons are shown in Tables 3 and 4.

Table 3: Matched comparisons of the four primary measures.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. <i>p</i>	Sign-flip <i>p</i>
Coauthor-citation rate	74,690	0.000	0.028	0.000	[0.000, 0.000]	0.048	[0.047, 0.050]	0.140	< 0.001	< 0.001
Reciprocity	39,441	0.000	0.000	0.000	[0.000, 0.000]	0.096	[0.092, 0.100]	0.341	< 0.001	< 0.001
Local clustering	102,626	0.000	0.000	0.000	[0.000, 0.000]	0.081	[0.079, 0.084]	0.264	< 0.001	< 0.001
Outgoing HHI	74,690	0.074	0.030	0.036	[0.035, 0.037]	0.097	[0.095, 0.099]	0.594	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. Median differences can be zero because many values are tied at zero, while mean differences show the average shift. Positive differences and positive rank-biserial effects indicate Case > Control. BH adjusts the four displayed *p*-values for multiple testing; the sign-flip *p*-value comes from randomly reversing pair directions.

Table 4: Matched comparisons of secondary measures.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. <i>p</i>	Sign-flip <i>p</i>
Self-citation rate	75,526	0.000	0.007	0.000	[0.000, 0.000]	0.033	[0.031, 0.034]	0.177	< 0.001	< 0.001
Journal endogamy	79,700	0.008	0.064	-0.027	[-0.027, -0.026]	-0.009	[-0.010, -0.007]	-0.235	< 0.001	< 0.001
Within-sample citation balance	59,924	0.000	-0.024	0.000	[0.000, 0.000]	-0.020	[-0.028, -0.011]	-0.018	< 0.001	< 0.001
Maximum annual dyadic surge share	75,526	0.123	0.069	0.047	[0.046, 0.048]	0.094	[0.092, 0.096]	0.526	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. Median differences can be zero because many values are tied at zero, while mean differences show the average shift. Positive differences and positive rank-biserial effects indicate Case > Control. BH adjusts the four displayed *p*-values for multiple testing; the sign-flip *p*-value comes from randomly reversing pair directions.

265 The four primary mean paired differences were Coauthor-citation rate 0.048, 95% bootstrap CI [0.047, 0.050]; Reciprocity 0.096, 95% bootstrap CI [0.092, 0.100]; Local clustering 0.081, 95% bootstrap CI [0.079, 0.084]; Outgoing HHI 0.097, 95% bootstrap CI [0.095, 0.099]. 3 median differences were zero. Outgoing HHI had a median paired difference of 0.036, 95% bootstrap CI [0.035, 0.037].

270 The median difference in same-journal referencing was -0.027, 95% CI [-0.027, -0.026]; the median difference in the maximum year-to-year increase to one recipient was 0.047, 95% CI [0.046, 0.048].

Table 5: Checks using exact h_5 matches and same-year collaborations.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. p	Sign-flip p
Coauthor-citation rate	47,128	0.035	0.025	0.000	[0.000, 0.000]	0.075	[0.072, 0.077]	0.296	< 0.001	< 0.001
Reciprocity	24,032	0.140	0.000	0.015	[0.000, 0.031]	0.147	[0.142, 0.153]	0.470	< 0.001	< 0.001
Local clustering	64,176	0.000	0.000	0.000	[0.000, 0.000]	0.091	[0.088, 0.094]	0.330	< 0.001	< 0.001
Outgoing HHI	47,128	0.071	0.030	0.034	[0.033, 0.035]	0.097	[0.095, 0.099]	0.574	< 0.001	< 0.001
Coauthor-citation rate (same-year included)	74,690	0.068	0.080	0.000	[0.000, 0.000]	0.051	[0.049, 0.053]	0.105	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. The first four rows use exact- h_5 matches and BH correction across the primary outcomes. The same-year coauthor row uses the full cohort as a separate one-outcome check. A confidence interval (CI) gives the range supported by the paired bootstrap. Median differences can be zero because many values are tied at zero, while mean differences show the average shift.

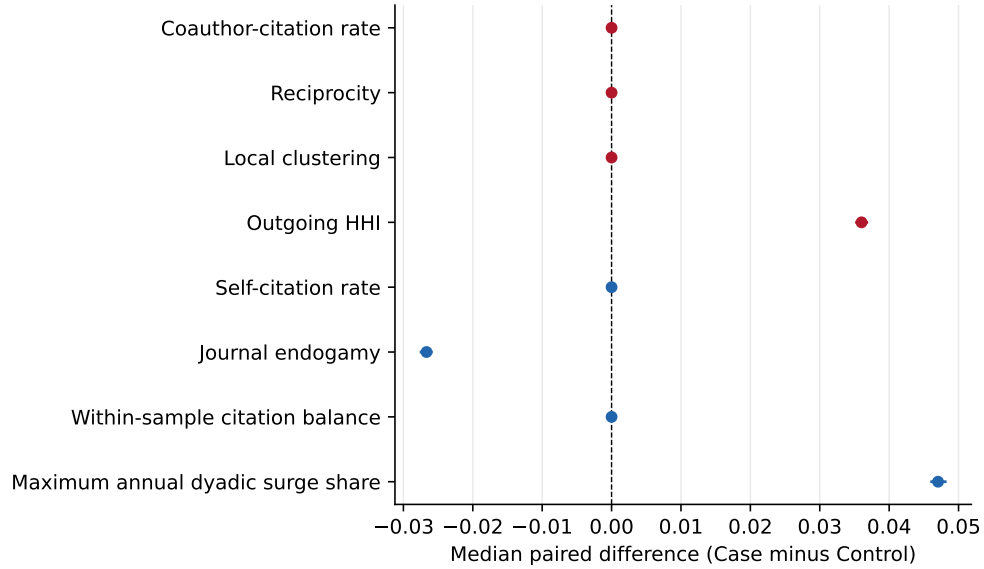


Figure 2: Matched effect estimates by outcome. Points show median paired differences and bars show paired-bootstrap 95% confidence intervals; mean differences and mean intervals are reported in Tables 3 and 4.

Pair counts are reported separately because the missingness pattern differs by measure. The count for every metric is checked not to exceed the 102,626 primary pairs.

Table 6: Screen results by journal-impact tier.

Tier	Eligible	Screenable	Flagged	Flagged share	Case/Control enrichment	Case share of flags	Fisher exact p
Case	75,954	53,365	1,880	2.48%	15.26	92.07%	< 0.001
Control	99,889	71,146	162	0.16%	15.26	–	< 0.001

Eligibility requires identified non-self outgoing citation weight. Enrichment is the Case flagged share divided by the Control flagged share. The Fisher exact p -value tests whether the two flagged shares differ. Flags are screening results and do not establish intent or misconduct.

Table 7: The two screen rules and their overlap.

Detector exceeds threshold	Cohesion rule: no	Cohesion rule: yes	Total
No	118,525 (95.19%)	14 (0.01%)	118,539 (95.20%)
Yes	3,930 (3.16%)	2,042 (1.64%)	5,972 (4.80%)
Total	122,455 (98.35%)	2,056 (1.65%)	124,511 (100.00%)

Cells are counts (percent of 124,511 screenable author–subject rows). Spearman’s ρ between detector score and cohesion exceedance count was 0.421 ($n = 124,511$; $p < 0.001$). The four cohesion measures are detector inputs, so this is an overlapping interpretable restriction rather than independent corroboration.

275 8.2 Sensitivity of primary comparisons

The exact- h_5 check had positive mean differences for all four primary comparisons; outgoing HHI had a median difference of 0.034 (95% CI [0.033, 0.035]). Using the same-year coauthor definition, the median difference was 0.000 (95% CI [0.000, 0.000]).

8.3 Control-referenced anomaly enrichment

280 The screen flagged 1880 Cases and 162 Controls (2.48% versus 0.16%; 15.26-fold difference; Fisher exact $p < 0.001$).

The two screen rules agreed for 2,042 rows; 3,930 passed only the first rule and 14 only the second. Their scores were positively related ($\rho = 0.421$, $p < 0.001$).

Removing one input at a time changed little: flag overlap ranged from 0.931 to 0.995,
285 and Case/Control enrichment ranged from 14.83 to 15.58. Details appear in Table 12.

Table 8: Sensitivity of screen results.

Control percentile	Flagged count: median [range]	Case share	Control share	Enrichment	Overlap with main screen
97.5%	3968 [3942–3985]	4.67%	0.42%	11.09	0.51
99.0%	2046 [2029–2052]	2.48%	0.16%	15.25	1.00
99.5%	1213 [1180–1224]	1.50%	0.07%	20.46	0.59

Each row summarises ten fixed seeds; the main screen uses the 99th percentile and seed 42. Enrichment is the Case flagged share divided by the Control flagged share. Overlap is the Jaccard share of flagged rows retained by both screens.

Table 9: Where matched authors cite.

Citing group	Cited Case	Cited Control
Case	27500.67	1765.51
Control	431.67	7104.63

Cells are cumulative fractional citation weights. The within-tier share is 94.03%, and weighted assortativity is 0.828; this compares same-tier weight with the amount expected from the row and column totals. The two-sided 10,000 within-pair tier-label-swap p-value is < 0.001 .

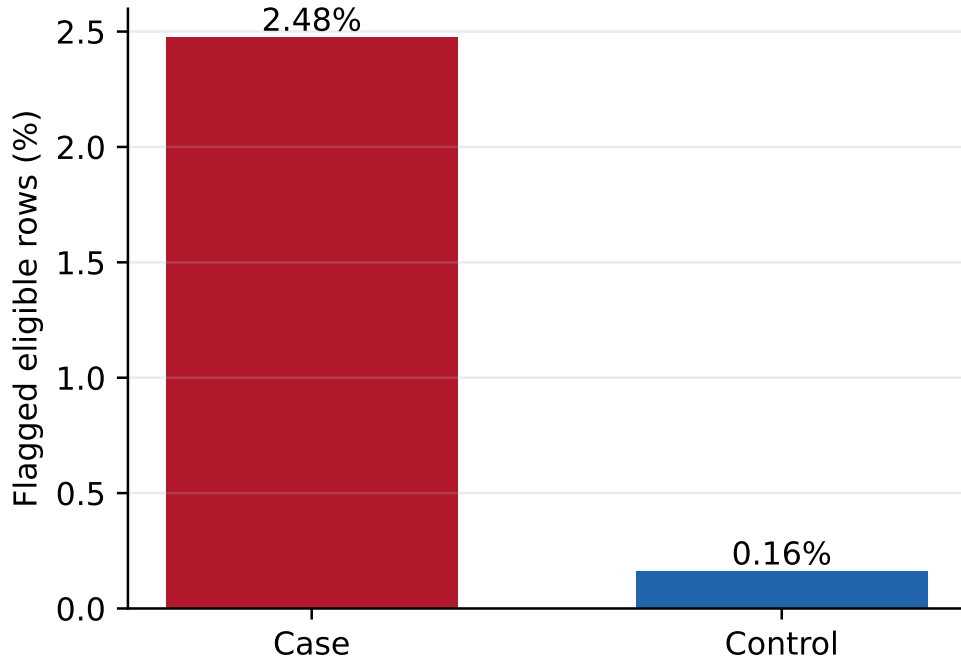


Figure 3: Share of eligible author–subjects flagged in each group under the declared screen. Values and supporting counts are generated from the same final flag used in all downstream analyses.

8.4 Fractional group mixing

94.03% of matched citation weight stayed within the same group (label-swap $p < 0.001$).

Table 10: Matched reciprocal-clique membership comparison.

Primary rule	All groups	Recip. groups	Case members	Control members	Paired diff.	Case only	Control only	Exact p
$k \geq 4, r \geq 0.50$	36,311	9,362	11,252	3,925 (3.82%)	7.14 pp	10,678	3,351	< 0.001
(10.96%)								

A candidate group is a maximal group in which every pair has a citation in at least one direction. The primary rule requires at least four members, density at least 0.75, and weighted reciprocity at least 0.50. Authors in overlapping qualifying groups are counted once per subject. The exact two-sided matched test uses only discordant pairs.

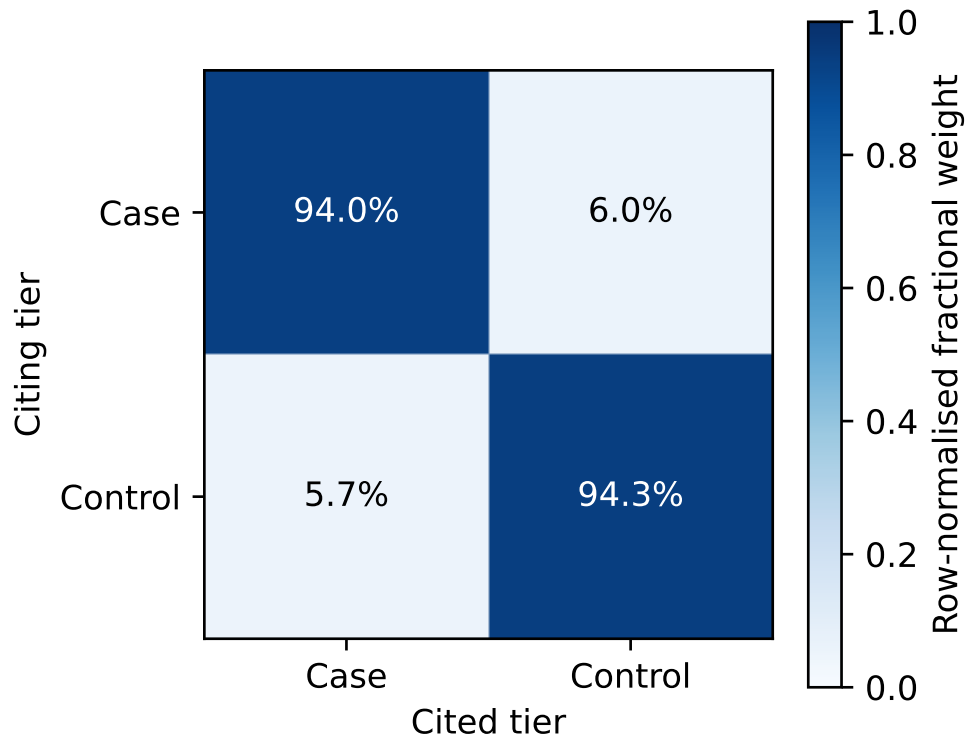


Figure 4: Fractional citation-weight mixing between Case and Control nodes. The comparison distribution is based on within-pair group-label swaps.

8.5 Formal reciprocal-clique structure

Primary-rule reciprocal-clique membership was 10.96% for Cases and 3.82% for Controls
290 (paired difference 7.14 percentage points; exact $p < 0.001$).

Table 11: Sensitivity of the clique rule.

Minimum group size	Recip. threshold	Recip. groups	Case rate	Control rate	Paired diff.	Case only	Control only	Exact p	Subjects + / - / =
3	0.25	33,041	21.11%	9.89%	11.22 pp	19,007	7,491	< 0.001	5/0/0
3	0.50	10,237	12.21%	4.82%	7.39 pp	11,770	4,190	< 0.001	5/0/0
3	0.75	1,420	2.85%	1.25%	1.60 pp	2,879	1,238	< 0.001	5/0/0
4	0.25	31,621	19.47%	8.21%	11.26 pp	17,865	6,311	< 0.001	5/0/0
4	0.50	9,362	10.96%	3.82%	7.14 pp	10,678	3,351	< 0.001	5/0/0
4	0.75	982	2.05%	0.82%	1.22 pp	2,080	823	< 0.001	4/1/0
5	0.25	25,836	14.62%	3.99%	10.63 pp	14,145	3,238	< 0.001	4/0/1
5	0.50	6,677	8.03%	1.82%	6.21 pp	8,020	1,648	< 0.001	4/0/1
5	0.75	443	1.18%	0.35%	0.83 pp	1,213	361	< 0.001	3/1/1

The directed density threshold is fixed at 0.75. The reciprocity threshold is the minimum share of two-way citation weight. Membership is unique by author–subject, and the last column reports subjects with higher Case rates, higher Control rates, or ties.

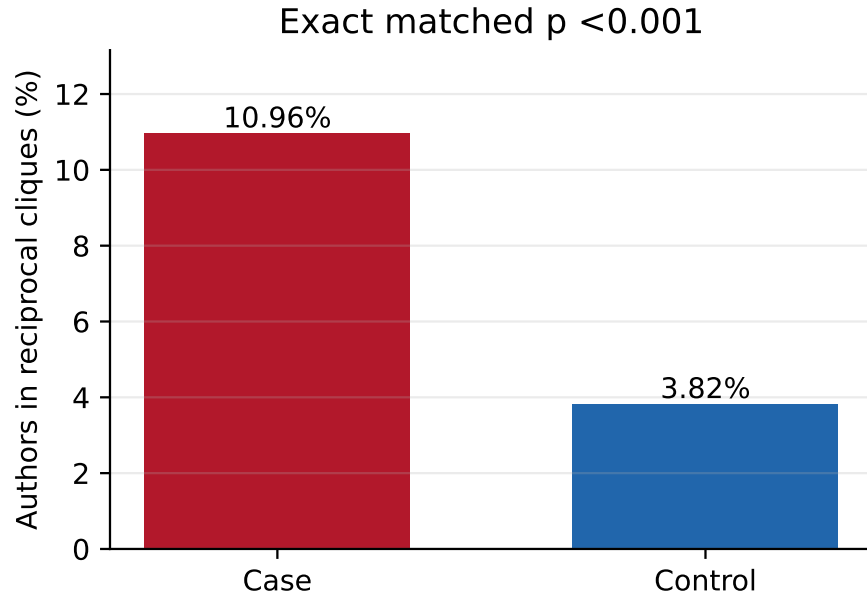


Figure 5: Unique author–subject membership rates in primary-rule reciprocal citation cliques. Authors in overlapping cliques are counted once.

Here, a clique is a group of at least four matched authors in which every pair is connected by a citation in at least one direction. The primary inference compares the binary author–subject membership indicator within pairs rather than counting repeated appearances in overlapping cliques. There were 10678 pairs in which only the Case belonged and 3351 in which only the Control belonged; the exact matched result was $p < 0.001$. The subject-level 295 breakdown is reported with every threshold sensitivity.

9 Discussion

This study compares two journal-impact groups after matching authors within subject and on recent citation productivity. The result is an association within matched pairs. It is
300 not an estimate of what would happen if an author changed venues, because venue choice, collaboration, topic, and citation practice can affect one another.

The four primary mean paired differences were Coauthor-citation rate 0.048, 95% bootstrap CI [0.047, 0.050]; Reciprocity 0.096, 95% bootstrap CI [0.092, 0.100]; Local clustering 0.081, 95% bootstrap CI [0.079, 0.084]; Outgoing HHI 0.097, 95% bootstrap CI [0.095, 0.099]. 3
305 median differences were zero. Outgoing HHI had a median paired difference of 0.036, 95% bootstrap CI [0.035, 0.037]. This comparison complements the author-level concentration measures of Evdaimon et al., while measuring the distribution of references sent by an author rather than the number of authors needed to account for received citations (Evdaimon, Ioannidis, et al., 2024).

310 The secondary results add detail. The median difference in same-journal referencing was -0.027, 95% CI [-0.027, -0.026]; the median difference in the maximum year-to-year increase to one recipient was 0.047, 95% CI [0.046, 0.048]. These measures describe different parts of citation behavior from the journal-level citation stacking studied by Heneberg (Heneberg, 2016). The screen flagged 2.48% of Case rows and 0.16% of Control rows, a 15.26-fold
315 difference. 94.03% of matched citation weight stayed within the same group (label-swap $p < 0.001$). These are descriptive results that show how the groups differ; they do not identify a cause.

The clique analysis makes the main pattern more specific. Primary-rule reciprocal-clique membership was 10.96% for Cases and 3.82% for Controls (paired difference 7.14 percentage
320 points; exact $p < 0.001$). Membership is defined once per author–subject even when cliques overlap, and the exact test preserves the matched design. The evidence therefore concerns which matched authors belong to the defined structure, while leaving the reasons for the pattern open.

Kojaku et al. detect anomalous groups of journals by comparing journal-level citation structure with network expectations (Kojaku et al., 2021). Our result instead compares author-325 level clique membership within matched pairs under a declared structural rule. The approaches share a relational perspective, but their populations, estimands, and null comparisons differ, so their estimates are not numerically interchangeable and neither structural result alone establishes misconduct.

The screen and clique rule are measurement tools for follow-up. They summarize citation 330 links and cannot determine whether a citation was appropriate, coordinated, or intentional. Narrow specialties, stable research teams, and institutional or career-stage differences can also create cohesive citation groups.

10 Limitations

The estimates apply to the fixed 2020–2024 snapshot and the author–subject observations 335 that meet the stated inclusion criteria. The graph metrics therefore describe the defined analytic population and graph construction.

The five broad subject categories cannot capture fine topical proximity. The journal classifier was not independently validated against blinded human coding, and the publication-weighted Eigenfactor cutoffs remain sensitive to the observed journal network. ORCID 340 availability, the 70% portfolio rule, and the positive- h_5 matching pool define the study population. Matching on subject-specific h_5 improves comparability within that pool but does not balance geography, language, institution, team structure, career age, or unmeasured research topic.

Graph metrics depend on population boundaries. Ego-layer measures include identified 345 recipients outside the matched cohort, whereas reciprocity, clustering, balance, and mixing are restricted to matched nodes. Results from the latter should not be interpreted as full-network properties. The 2020–2024 citation window limits observed citation behavior, while historical

work and author metadata in the snapshot can left-censor or misdate prior collaborations.

350 The year dimension of the edge data supports coauthor timing and the scalar surge-share outcome, but is not analyzed as a group-specific trajectory or group-by-year interaction; we therefore make no claim that a Case–Control gap widens over time.

The anomaly thresholds and Isolation Forest define an operational screen, not a validated diagnostic. Percentile, seed, and leave-one-feature-out sensitivity quantify some model
355 dependence but do not supply ground-truth labels or make the overlapping restriction independent. Finally, structural metadata cannot reveal the textual function, scientific relevance, or motivation of a citation (J. Liu et al., 2024).

11 Conclusion

All four primary mean differences favored Cases; all four comparisons were significant after
360 adjustment, and 3 median differences were zero.

Lower-impact-group authors were more likely than matched higher-impact-group authors to belong to formally defined reciprocal citation cliques. The paired membership difference was 7.14 percentage points (exact $p < 0.001$). This is an associational result for the defined cohort and supplies a pattern for follow-up without assigning a cause or intent.

365 12 Author Contributions

Conceptualization: P.-A.S., D.S.; Methodology: P.-A.S.; Software: P.-A.S., G.A.; Data curation: P.-A.S., G.A.; Validation: P.-A.S., G.A., D.S.; Formal analysis: P.-A.S., G.A.; Visualization: P.-A.S.; Writing, original draft: G.A.; Writing, review and editing: P.-A.S., G.A., D.S.; Supervision: D.S.; Project administration: D.S.

13 Competing Interests

370

The authors declare no competing interests.

14 Funding Information

This research did not receive a specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

15 Data and Code Availability

375

The v4.0.0 software and aggregate replication materials for this revision are archived on Zenodo (Spanakis et al., 2026). The fixed raw snapshot is not redistributed.

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A Upstream Venue-Score Specification

For subject s , let $Z_{jk}^{(s)}$ count citations from citing journal j to cited journal k . The diagonal is removed, and each non-dangling citing-journal column of H_s is

$$(H_s)_{kj} = \frac{Z_{jk}^{(s)}}{\sum_{\ell} Z_{j\ell}^{(s)}}. \quad (9)$$

Let a_s be the normalized journal article-count vector and d_s indicate dangling columns. With
 445 damping $\alpha = 0.85$, the subject-specific influence vector is obtained by

$$\pi_s = \alpha H_s \pi_s + \left[\alpha d_s^\top \pi_s + (1 - \alpha) \right] a_s. \quad (10)$$

The received-influence vector $H_s \pi_s$ is normalized to sum to 100 within subject. Thus citation mass flows from citing to cited journals, journal self-citations are excluded, and dangling mass and restart mass follow article counts. These fixed journal-level scores are upstream inputs to the author matching and are not author-network outcomes.

450 B Matching and Sensitivity Specifications

Cases and Controls require a non-empty ORCID, at least three eligible works, and a 70% portfolio share in the relevant journal group. The matching pool additionally requires a positive h_5 value; zero- h_5 profiles are not candidates. Within that pool, candidate eligibility uses the inclusive hard caliper $|h_{5,\text{Case}} - h_{5,\text{Control}}| \leq 3$. Greedy selection proceeds without
 455 replacement within subject in the total order subject, ascending absolute h_5 distance, Case ORCID, and Control ORCID. The generated cohort must pass the same subject-keyed positive- h_5 join and caliper checks. Exact- h_5 pairs form a sensitivity subset rather than a replacement cohort.

The paired bootstrap resamples pair identifiers with replacement. The sign-flip test acts
 460 on complete-pair differences for one metric at a time. Multiple-testing adjustment is not

pooled across primary and secondary families. The same-year coauthor sensitivity changes only $c_{ij} < t$ to $c_{ij} \leq t$; all other definitions and pair membership remain fixed.

C Anomaly-Screen Specification

Controls define subject-specific robust centers, scales, detector fits, score cutoffs, and univariate cohesion cutoffs. Cases do not affect those reference quantities. The canonical feature output 465 retains the detector score, cohesion-restriction indicator, citation eligibility, and final flag for each (ORCID, s , tier) row. The downstream mixing routine consumes that final flag rather than recomputing or substituting a label.

The primary screen uses seed 42 and the 99th Control percentiles. The threshold analysis repeats the complete fit at the 97.5th, 99th, and 99.5th percentiles for ten fixed seeds. A row 470 with an unavailable required detector input is not zero-imputed; it receives no detector score, cannot be flagged, and is identified by a separate detector-completeness field. The overlap table is restricted to rows on which both criteria are defined.

The anomaly score is the negative of the library's `score_samples` output. The library's `contamination=auto` decision cutoff is not used; the reported cutoff is computed from the 475 subject-specific Control scores.

Table 12: Screen stability when one input is removed.

Detector feature omitted	Flags	Case flags	Control flags	Enrichment	Overlap with main screen
Self-citation rate	2,056	1,889 (2.49%)	167 (0.17%)	14.88	0.993
Coauthor-citation rate	1,901	1,753 (2.31%)	148 (0.15%)	15.58	0.931
Reciprocity	2,052	1,886 (2.48%)	166 (0.17%)	14.94	0.995
Local clustering	2,050	1,883 (2.48%)	167 (0.17%)	14.83	0.990
Outgoing HHI	1,907	1,758 (2.31%)	149 (0.15%)	15.52	0.934

Each fit uses the 99.0% Control percentile and seed 42. The main detector-complete population and cohesion restriction are held fixed; Case and Control percentages retain eligible-row denominators. Overlap is the Jaccard share of flagged rows retained by both screens.

D Reproducibility Checks

The rebuild checks the following manuscript-facing invariants:

- (ORCID, s , tier) is unique in the feature output;
- 480 • generated pairs have complete subject-keyed h_5 records, satisfy the inclusive three-point caliper, and do not reuse an author–subject key;
- every analytic edge has non-null endpoints and is unique on (s, i, j, t) ;
- each per-metric complete-pair count is no greater than 102,626;
- every observed surge share lies in $[0, 1]$;
- 485 • the same final flag supplies every flag-dependent enrichment and figure artifact; and
- reported numerical values enter through generated LaTeX artifacts rather than hand-copied prose.

SQL fixtures separately test fractional weight, null endpoints, work-reference deduplication, subject retention, and prior-coauthor timing. Python unit tests cover full-ORCID cohort
 490 construction, matching eligibility and deterministic order, dynamic cohort validation, metric formulas, paired inference, unique clique membership, exact matched clique inference, detector overlap, feature ablation, weighted mixing, and flag reuse.

E Journal Subject Classification

The retained journal mapping was produced with OpenAI `gpt-4.1` (snapshot `gpt-4.1-2025-04-14`) at zero temperature from journal names and representative work titles and abstracts.⁴⁹⁵ Prompts and cached outputs are part of the replication materials. Because the mapping has no independent human-validation sample and no open-weight baseline, classification error remains a limitation rather than a measured source of uncertainty (LLM Guidelines Consortium, 2024).